A Distributed Risk-Aware Bidding Strategy for Incorporating Renewable Generation into Real-Time Electricity Market

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Abstract—This paper proposes a risk-aware optimization framework in which firms' sensitivity to uncertainty serves as a tunable coordination variable. We show that, for a class of stochastic market games, there exists a structured relationship between firms' risk parameters and the resulting Nash equilibrium, such that appropriately chosen risk sensitivities induce Nash equilibria that lie on the Pareto frontier. Based on this insight, we develop a distributed algorithm that allows firms to estimate required aggregate quantities and adapt their risk parameters using only local communication and minimal information exchange. The proposed approach preserves the decentralized nature of the market: firms remain self-interested in their primary decisions, resulting in Pareto-optimal Nash equilibria, and convergence of the distributed algorithm follows from consensus theory under mild connectivity conditions.

Keywords: Risk-aware bidding, Nash equilibrium, Pareto efficiency, stochastic market, distributed optimization, renewable energy

I. Introduction

Competitive electricity markets operating under uncertainty exhibit a fundamental tension between individual rationality and collective efficiency. Renewable energy operators in particular face this challenge acutely: their generation is stochastic and correlated through shared weather conditions, their output forecasts are uncertain, and their bidding decisions must be submitted before real-time prices are known [1], [2], [3]. When each operator optimizes its own expected profit independently, the resulting Nash equilibrium generally fails to achieve socially optimal outcomes, even when all firms behave rationally [4], [5]. This gap between Nash and Pareto-optimal performance is not merely a theoretical artifact—in markets with many correlated renewable operators, it translates directly into foregone collective profit and inefficient price signals [6], [7].

Existing studies on stochastic Cournot and oligopoly games have examined the impact of uncertainty and risk preferences on equilibrium outcomes, but typically treat risk attitudes as fixed exogenous parameters [8], [9], [10]. Under this assumption, the equilibrium is determined once risk preferences are assigned, leaving no room to steer outcomes. Separately, Pareto efficiency has been studied through centralized optimization or cooperative formulations that require full information sharing [11], [12], [13]. These approaches are impractical in deregulated markets where operators are self-interested and unwilling to disclose private cost and forecast data. The key question that remains unaddressed is whether Nash equilibrium behavior can be reconciled with Pareto-optimal outcomes without centralized coordination or private information disclosure.

This paper answers that question affirmatively by identifying risk sensitivity itself as the coordination variable. We show that, for a class of stochastic renewable energy market games, there exists a structured relationship between the operators' risk parameters and the resulting Nash equilibrium bidding strategies, such that appropriately chosen risk levels induce Nash equilibria that lie on the Pareto frontier. In this sense, risk is not merely a preference parameter but a mechanism through which self-interested agents can be guided toward efficient collective outcomes without violating incentive compatibility. This perspective connects to the mechanism design literature [14], [15]: rather than redesigning the market structure, the utility prescribes risk parameters that act as a voluntary coordination instrument, inducing Pareto-efficient Nash behavior without coercive transfers or centralized side payments.

The key enabler for a distributed implementation is that the Pareto condition depends only on the aggregate risk level across operators, not on individual bidding strategies directly. This means coordination reduces to reaching consensus on a single scalar quantity, which can be accomplished through local communication without revealing private information. Building on this insight, we develop a distributed consensus-based algorithm that allows each operator to estimate the required aggregate quantities and set its risk parameter using only neighborhood communication [16], [17]. Analytical results establish existence and explicit characterization of Pareto-optimal Nash equilibria, and convergence of the distributed algorithm is established under a mild connectivity condition. A numerical example illustrates how risk-aware adaptation shifts equilibrium behavior from inefficient Nash outcomes to Pareto-optimal payoffs, and a sensitivity analysis quantifies the performance impact of physical bid constraints.

The remainder of the paper is organized as follows. Section II formulates the stochastic market model and risk-aware objectives. Section III characterizes the Nash equilibrium under fixed risk parameters. Section IV establishes conditions under which Nash equilibria achieve Pareto efficiency through structured risk allocation. Section V presents a distributed algorithm based on consensus to realize these outcomes. Numerical results are provided in Section VI, followed by concluding remarks in Section VII.

II. Problem Formulation

In this paper, we consider the simple case that all traditional generation assets are owned by a single utility company[1] while renewable generation is operated by individual entities (one of which could be the utility company). And, the utility company and the renewable generation operators are the participants in both day-ahead and real-time electricity markets for profit.

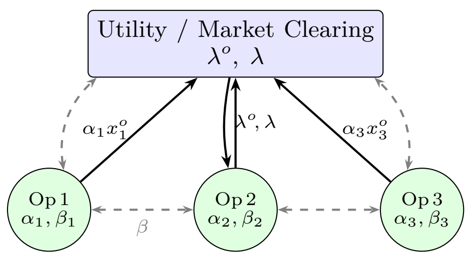

Fig. 1. System and communication structure. Solid: bid submission and price return. Dashed: consensus links for Algorithm 1.

In an electricity market, utility aims to optimize a composite objective function in the form of

$$J(P_g) = a_2 P_g^2 + a_1 P_g + a_0, \tag{1}$$

where a_i are cost coefficients, and P_g is the aggregated power of traditional generation. For simplicity, marginal costs of operating renewables and battery storage sites are modeled as zero, and hence they need not to be included explicitly in objective function (1).

Let n denote the total number of renewable generation operators and $\Omega = \{1, \cdots, n\}$ be the index set. For the day-ahead market, their renewable generations, P_j for $j \in \Omega$, are modeled as stochastic variables, and similarly the grid total load P_L is also modeled as a stochastic variable. Let us form the whole vector of random variables as

$$x = \begin{bmatrix} P_1 & \cdots & P_n & P_L \end{bmatrix}^T. \tag{2}$$

We assume that probability distribution of random vector x is Gaussian as follows:

$$f(x) = (2\pi)^{-\frac{n+1}{2}} \Sigma^{-\frac{1}{2}} e^{-\frac{1}{2}(x-x^o)^T \Sigma^{-1}(x-x^o)},$$

where $x^o = \mathbb{E}[x]$ is the vector of forecast (for the day-ahead market), and

$$\begin{aligned} \Sigma &= \mathbb{E}[(x-x^o)(x-x^o)^T] \\ &= \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} & \sigma_{1L} \\ \vdots & \cdots & \vdots & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} & \sigma_{nL} \\ \sigma_{1L} & \cdots & \sigma_{nL} & \sigma_{LL} \end{bmatrix}, \end{aligned}$$

is the covariance matrix with $\sigma_{ij} = \sigma_{ji}$. In the covariance matrix, the last row and column correspond to covariances between the renewable generations and the load while $\sigma_{LL} = \sigma_L^2 > 0$ is the variance of the load. It is known that $\sigma_{ii} = \sigma_i^2 > 0$, $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$, $\sigma_{iL} = \rho_{iL}\sigma_i\sigma_L$, where $\rho_{ij}, \rho_{iL} \in [0, 1]$. Typically, $\rho_{ij}, \rho_{iL} \neq 0$ because renewable generations and load are correlated through the projected weather condition.

The Gaussian model is adopted for notational consistency with standard stochastic Cournot frameworks, but the key results of this paper require only that x have finite mean x^o and covariance Σ. The risk penalty $\mathbb{E}[(\lambda - \lambda^o)^2]$ and total profit Π both expand as polynomials in the bidding coefficients with moment-based coefficients determined entirely by x^o and Σ (see (14) and (15)), so Theorems 1–2 and Corollaries 1–5b hold for any joint distribution with finite second moments, including Weibull wind and beta solar distributions common in practice. Gaussianity would be needed only for distributional risk measures beyond variance (e.g., CVaR) or for characterizing higher moments of individual profits, which are deferred to future work.

The day ahead market clears the projected aggregated power generation P_g^o and its associated price λ^o by minimizing $J(P_g^o)$ under the following power balance constraint:

$$P_g^o + \sum_{j \in \Omega} \alpha_j x_j^o = x_{n+1}^o, \tag{3}$$

where $x_{n+1}^o = P_L^o$, $\alpha_j \in [0, 1]$ if no overbidding is allowed or $\alpha_j \in [0, \infty)$ if overbidding is permitted, and $\alpha_j x_j^o = \alpha_j P_j^o$ represent the bids by the renewable generation operators into the day-ahead market. Similarly, the real-time market clears the aggregated power generation P_g and its associated price λ by minimizing $J(P_g)$ under the following real-time power balance constraint:

$$P_g + \sum_{j \in \Omega} x_j = x_{n+1}. \tag{4}$$

It is well known that, with global information, day-ahead price λ^o and real-time λ are the dual variables associated with constraints (3) and (4), respectively.

[1]The proposed results apply to the general case of multiple utility companies of individual cost functions except for notational complexity.

The renewable generation operators aim to maximize the following objectives[2] individually:

$$J_i = \Pi_i - \beta_i \mathbb{E}[(\lambda - \lambda^o)^2], \tag{5a}$$

where $\beta_i \in \mathbb{R}$,

$$\Pi_i = \lambda^o \alpha_i x_i^o + \mathbb{E}[\lambda(x_i - \alpha_i x_i^o)], \tag{5b}$$

and the pair $\{\alpha_i,\ \beta_i\}$ represents the bidding strategy of the i-th operator into the electricity markets. Throughout this paper, J_i exclusively denotes the risk-aware objective defined in (5a); the special case $\beta_i = 0$ reduces J_i to pure expected profit maximization and is referred to as the risk-neutral strategy. If $\beta_i = 0$ is chosen, the i-th entity aims to maximize its average profit from the total revenues from participating in the day ahead and real time markets. By selecting $\beta_i > 0$, the i-th entity also attempts to reduce its profit variation by suppressing variations of real time price λ around day ahead price λ^o. By choosing $\beta_i < 0$, the ith entity tries to maximize its profit by encouraging more variations of real time price λ around day ahead price λ^o. In other words, the i-th entity is risk averse if $\beta_i > 0$ and risk seeking if $\beta_i < 0$.

The risk-aware bidding strategy $\{\alpha_i, \beta_i\}$ offers four advantages: (i) achieving Pareto-optimal Nash equilibria that maximize the total profit $\sum_{i\in\Omega} \Pi_i$; (ii) enabling distributed implementation without centralized coordination; (iii) maintaining incentive compatibility as firms remain self-interested; and (iv) preserving privacy through minimal information exchange. Power grid congestion is not considered in the paper but can be considered.

III. Centralized Risk-Aware Solution Under Global Information

The electricity markets are bi-level optimization problems linked together by dual variables. At the upper level, they are quadratic optimization problems with equality constraints, that is, the markets perform their economic dispatch with the following Lagrangians:

$$L^o = a_2 (P_g^o)^2 + a_1 P_g^o + a_0 + \lambda^o \left(x_{n+1}^o - P_g^o - \sum_{j\in\Omega} \alpha_j x_j^o \right), \tag{6a}$$

$$L = a_2 P_g^2 + a_1 P_g + a_0 + \lambda \left(x_{n+1} - P_g - \sum_{j\in\Omega} x_j \right). \tag{6b}$$

Applying the first-order optimality condition to the Lagrangians yields the following solutions for the prices:

$$\lambda^o = 2a_2 P_g^o + a_1 = 2a_2 \left(x_{n+1}^o - \sum_{j\in\Omega} \alpha_j x_j^o \right) + a_1, \tag{7a}$$

$$\lambda = 2a_2 P_g + a_1 = 2a_2 \left(x_{n+1} - \sum_{j\in\Omega} x_j \right) + a_1. \tag{7b}$$

The lower level optimization problems are also quadratic, and the centralized solutions under global information are summarized into the following lemma and theorem. Lemma 1 provides the solution of cooperative optimization when all the entities bind themselves together and act as one. In this case, the best outcome for the group is achieved under the risk neutral strategy.

Lemma 1: Consider the case that all renewable generation entities form a coalition by sharing all the information and setting $\alpha_i = \alpha$ and $\beta_i = \frac{1}{n}\beta$, respectively. Then, for any choice of $\beta \in [-(4a_2)^{-1},\ +\infty)$, the optimal bidding strategy to optimize objective function $\sum_i J_i$ with J_i in (5a) is

$$\alpha^* = \frac{1 + 4a_2\beta}{2(1 + 2a_2\beta)}. \tag{8}$$

In addition, by further choosing β_i to optimize $\sum_i \Pi_i$ with Π_i in (5b), the optimal bidding strategy $\{\alpha^*,\ \beta^*\}$ is given by

$$\alpha^* = \frac{1}{2}, \quad \beta^* = 0. \tag{9}$$

Proof: It follows from (5b) and (7a) that, if $\alpha_i = \alpha$,

$$\begin{aligned} \Pi_r &= \sum_{j\in\Omega} \Pi_j = \lambda^o \alpha x_r^o + \mathbb{E}[\lambda(x_r - \alpha x_r^o)] \\ \lambda^o &= 2a_2 \left(x_{n+1}^o - \alpha x_r^o \right) + a_1, \end{aligned}$$

where $x_r = \sum_{j\in\Omega} x_j$ is the total renewable generation, and $x_r^o = \sum_{j\in\Omega} x_j^o$. Substituting (7b) into the above equations yields

$$\begin{aligned} \Pi_r &= [2a_2 \left(x_{n+1}^o - \alpha x_r^o \right) + a_1] \alpha x_r^o \\ &\quad + \mathbb{E}[(2a_2 \left(x_{n+1} - x_r \right) + a_1)(x_r - \alpha x_r^o)] \\ &= 2a_2 \alpha (1-\alpha)(x_r^o)^2 + a_1 x_r^o + 2a_2 \mathbb{E}[\left(x_{n+1} x_r - x_r^2 \right)] \\ &= 2a_2 \alpha (1-\alpha)(x_r^o)^2 + a_1 x_r^o \\ &\quad + 2a_2 \left(x_{n+1}^o x_r^o + \sigma_{rL} - (x_r^o)^2 - \sigma_r^2 \right), \end{aligned} \tag{10}$$

where $\sigma_{rL} \triangleq \sum_{i\in\Omega} \sigma_{iL}$ is the total renewable-load cross-covariance, $\mathbf{1}$ is the n-dimensional column vector of 1s, and

$$\begin{aligned} \sigma_r^2 &= \mathbb{E}[(x_1 + \cdots + x_n - (x_1^o + \cdots + x_n^o))^2] \\ &= \mathbb{E}[\mathbf{1}^T (x_r - x_r^o)(x_r - x_r^o)^T \mathbf{1}] \\ &= \mathbf{1}^T \Sigma_r \mathbf{1} \ = \sum_{i\in\Omega} \sum_{j\in\Omega} \sigma_{ij}, \end{aligned}$$

where $x_r = [x_1, \ldots, x_n]^T$ is the n-vector of renewable generations and Σ_r is the upper-left $n \times n$ subblock of Σ containing only the renewable-to-renewable covariances

[2] Strictly speaking, we should use $J_i' = \Pi_i - \beta_i \mathbb{E}[(\Pi_i - \mathbb{E}[\Pi_i])^2]$ which also yields a second-order polynomial in α_i as does (5a) but involves calculations of higher moments. Accordingly, (5a) is adopted instead.

σ_{ij}, $i, j \in \{1, \ldots, n\}$. On the other hand, it follows from (5a), (7a), and (7b) that, if $\beta_i = \frac{1}{n}\beta$,

$$\begin{aligned} J_r &= \sum_{j\in\Omega} J_j \\ &= \Pi_r - \beta\mathbb{E}[2a_2(x_{n+1} - x_{n+1}^o - x_r + \alpha x_r^o)^2] \\ &= 2a_2\alpha(1-\alpha)(x_r^o)^2 - 4a_2^2\beta(1-\alpha)^2(x_r^o)^2 \\ &\quad +a_1x_r^o + 2a_2\left(x_{n+1}^o x_r^o + \sigma_{rL} - (x_r^o)^2 - \sigma_r^2\right) \\ &\quad -4a_2^2\beta(\sigma_L^2 + \sigma_r^2 - 2\sigma_{rL}). \end{aligned}$$

Using the above expression, taking the partial derivative of J_r with respect to α, and setting the partial derivative to zero yield optimal bidding strategy (8).

Substituting solution (8) into (10) and differentiating Π_r with respect to β yields

$$\frac{\partial \Pi_r}{\partial \beta} = -2a_2(x_r^o)^2 \frac{4a_2^2\beta}{(1+2a_2\beta)^3},$$

which yields the optimal bidding strategy (9). This completes the proof. □

It follows from (8) that

$$\lim_{\beta\to -1/(4a_2)} \alpha = 0, \quad \lim_{\beta\to 0} \alpha = 0.5, \quad \lim_{\beta\to +\infty} \alpha = 1.$$

Since $\alpha \in [0,1]$, the design parameter β translates an operator's risk tolerance level into a specific choice of α value. Although the specific value corresponding to the risk-neutral choice may differ, this translation is applicable to not only cooperative coalition game but also noncooperative games.

The following theorem provides the solution to noncooperative optimization in which each agent devises its strategy according to its own objective function J_i.

Theorem 1: For renewable energy generation operators, their oligopoly game always has a Nash equilibrium. Specifically, for any choice of $\beta_i \in [-(4a_2)^{-1}, +\infty)$, the optimal bidding strategy α_i^* to optimize objective J_i in (5a) forms a Nash equilibrium corresponding to the solution to the linear algebraic equation: denoting $\boldsymbol{\alpha}^* = [\alpha_1^* \ \cdots \ \alpha_n^*]^T$,

$$A\boldsymbol{\alpha}^* = B, \tag{11}$$

where $x_r^o = \sum_{j\in\Omega} x_j^o$, $\eta_i = x_i^o/x_r^o \geq 0$ with $\sum_{i\in\Omega}\eta_i = 1$, and

$$A = [A_{ij}], \quad A_{ij} = \left\{ \begin{array}{ll} 2(1+2a_2\beta_i)\eta_i & \text{if } i = j \\ (1+4a_2\beta_i)\eta_j & \text{if } i \neq j \end{array} \right.,$$

and

$$B = [B_i], \quad B_i = (1+4a_2\beta_i).$$

In addition, the total profit $\Pi = \sum_{i\in\Omega}\Pi_i$ can be maximized by choosing β_j to maximize the following function:

$$g(\boldsymbol{\alpha}^*) = f(\boldsymbol{\alpha}^*)\left[1 - f(\boldsymbol{\alpha}^*)\right](x_r^o)^2, \tag{12}$$

where

$$f(\boldsymbol{\alpha}) \triangleq \sum_{i\in\Omega} \alpha_i\eta_i. \tag{13}$$

Proof: It follows from (5b), (7a), and (7b) that

$$\begin{aligned} \Pi_i &= \left[2a_2\left(x_{n+1}^o - \sum_{j\in\Omega}\alpha_j x_j^o\right) + a_1\right]\alpha_i x_i^o \\ &\quad +\mathbb{E}\left[\left(2a_2\left(x_{n+1} - \sum_{j\in\Omega} x_j\right) + a_1\right)(x_i - \alpha_i x_i^o)\right] \\ &= 2a_2\sum_{j\in\Omega}\alpha_i(1-\alpha_j)x_j^o x_i^o + a_1x_i^o \\ &\quad +2a_2\mathbb{E}\left[\left(x_{n+1} - \sum_{j\in\Omega}x_j\right)x_i\right] \\ &= 2a_2\sum_{j\in\Omega}\alpha_i(1-\alpha_j)x_j^o x_i^o + a_1x_i^o \\ &\quad +2a_2\left(x_{n+1}^o x_i^o + \sigma_{iL} - \sum_{j\in\Omega}(x_j^o x_i^o + \sigma_{ij})\right), \end{aligned} \tag{14}$$

where the identity $\mathbb{E}[x_ix_j] = x_i^o x_j^o + \sigma_{ij}$ is used. On the other hand, it follows from (7a) and (7b) that

$$\begin{aligned} &\mathbb{E}[(\lambda - \lambda^o)^2] \\ &= 4a_2^2\mathbb{E}\left[\left(x_{n+1} - x_{n+1}^o - \sum_{j\in\Omega}(x_j - \alpha_j x_j^o)\right)^2\right] \\ &= 4a_2^2\left[\sigma_{LL} + 2\sum_{j\in\Omega}\sigma_{jL} + 2\sum_{j\in\Omega}\sum_{l\in\Omega}\sigma_{jl}\right. \\ &\quad \left. + \left(\sum_{j\in\Omega}(1-\alpha_j)x_j^o\right)^2\right], \end{aligned} \tag{15}$$

The expression of J_i defined in (5a) becomes explicit in terms of α_i by combining (14) and (15). Taking the partial derivatives of J_i with respect to α_i yields

$$\begin{aligned} \frac{\partial J_i}{\partial \alpha_i} &= 2a_2(1+4a_2\beta_i)\sum_{j\in\Omega}(1-\alpha_j)x_j^o x_i^o \\ &\quad -2a_2\alpha_i(x_i^o)^2, \end{aligned} \tag{16a}$$

$$\frac{\partial J_i^2}{\partial \alpha_i^2} = -4a_2(1+2a_2\beta_i)(x_i^o)^2 < 0 \quad \forall \beta_i \geq -\frac{1}{4a_2}. \tag{16b}$$

Equation (16b) shows that, for each fixed choice of the other players' bids, J_i is strictly concave in α_i. Hence the first-order condition characterizes the unique best response of operator i. Setting (16a) to zero yields

$$(1+4a_2\beta_i)\sum_{j\in\Omega}(1-\alpha_j)x_j^o - \alpha_i x_i^o = 0, \tag{17}$$

and collecting these best-response equations for all i yields matrix equation (11), whose solution gives the Nash equilibrium.

It follows from (11) that α_i^* are functions of β_j. We know from (14) that

$$\begin{aligned}\Pi &= \sum_i \Pi_i \\ &= 2a_2\left[g(\boldsymbol{\alpha}) - (x_r^o)^2 + x_{n+1}^o x_r^o\right] + a_1 x_r^o \\ &\quad +2a_2\left[\sum_{i\in\Omega}\sigma_{iL} - \sum_{i\in\Omega}\sum_{j\in\Omega}\sigma_{ij}\right], \end{aligned} \tag{18}$$

where

$$g(\boldsymbol{\alpha}) \triangleq \sum_{i\in\Omega}\alpha_i x_i^o \sum_{j\in\Omega}(1-\alpha_j)x_j^o.$$

Hence, Π is maximized if function $g(\cdot)$ is maximized by choosing β_j. Simplifying the above expression of $g(\cdot)$ yields (12). This completes the proof. □

Theorem 1 extends the results in [18] in several ways. First, load uncertainties and correlations among the renewable generations are considered. As a result, we know from (18) that the degradation of total profit function Π due to uncertainties is

$$\Delta\Pi = 2a_2\left[\sum_{i\in\Omega}\sum_{j\in\Omega}\sigma_{ij} - \sum_{i\in\Omega}\sigma_{iL}\right], \tag{19}$$

is less as long as there is some correlation (e.g., due to weather) between renewable generation and load variation. Second, risk aware bidding strategies are considered, specifically, theorem 1 allows us to study and compare three classes of bidding strategies: risk neutral, risk adverse, and risk seeking. The latter two correspond to optimization of design parameters β_j.

Theorem 1 shows that the Nash equilibrium in terms of design parameters β_j can be further optimized, and the optimal choices of β_j are the main subject of this paper. To illustrate some of the specifics, let us first consider the two simple cases: $n = 1$ and $n = 2$. And the corresponding results are provided by the subsequent corollaries, followed by the risk neutral strategy for the general case of n operators.

Corollary 1: If $n = 1$, the optimal coalition (C) bidding strategy becomes the same as that in lemma 1 and with $g^C(\alpha^*) = \frac{1}{4}(x_r^o)^2$.

Corollary 2: In the case of duopoly ($n = 2$), the Nash bidding strategies with respect to J_i are:

$$\alpha_1^* = \frac{(1+4a_2\beta_1)}{3+4a_2(\beta_1+\beta_2)}\eta_1^{-1}, \tag{20a}$$

$$\alpha_2^* = \frac{(1+4a_2\beta_2)}{3+4a_2(\beta_1+\beta_2)}\eta_2^{-1}, \tag{20b}$$

Furthermore, β_i can be optimized with respect to $(\Pi_1 + \Pi_2)$, and the outcomes are strategically dependent as follows.

- Risk neutral (RN) optimal bidding strategy is $\beta_i^{NR} = 0$ for $i = 1, 2$; and the corresponding optimized profit is:
$$g^{RN}(\alpha_1^*, \alpha_2^*) = \frac{2}{9}(x_r^o)^2. \tag{21}$$
- Risk seeking (RS) optimal bidding strategy and the corresponding enhanced optimal profit are:
$$\beta_1^{RS} + \beta_2^{RS} = -\frac{1}{4a_2}, \tag{22a}$$
$$g^{RS}(\alpha_1^*, \alpha_2^*) = \frac{1}{4}(x_r^o)^2 = g^C(\alpha^*) > g^{RN}(\alpha_1^*, \alpha_2^*). \tag{22b}$$

Proof: It follows from (11) that

$$\begin{aligned}&\begin{bmatrix}2(1+2a_2\beta_1)\eta_1 & (1+4a_2\beta_1)\eta_2 \\ (1+4a_2\beta_2)\eta_1 & 2(1+2a_2\beta_2)\eta_2\end{bmatrix}\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} \\ &= \begin{bmatrix}(1+4a_2\beta_1) \\ (1+4a_2\beta_2)\end{bmatrix},\end{aligned}$$

whose solution is given by (20).

It follows from (20) that, when $\beta_i = 0$,

$$\alpha_1^* = \frac{1}{3}\eta_1^{-1}, \quad \alpha_2^* = \frac{1}{3}\eta_2^{-1}.$$

Substituting the above solutions into (13) and (12) yields (21).

To further optimize the risk-aware bidding strategy (20), we rewrite the solution as

$$\alpha_1^* = \theta_1\eta_1^{-1}, \quad \alpha_2^* = \theta_2\eta_2^{-1},$$

where

$$\theta_i = \frac{(1+4a_2\beta_i)}{3+4a_2(\beta_1+\beta_2)}. \tag{23}$$

It follows from (13) and (12) that

$$g(\alpha_1^*, \alpha_2^*) \quad = \quad (\theta_1+\theta_2)[1-(\theta_1+\theta_2)](x_r^o)^2,$$

which is maximized to be that in (22b) if

$$\theta_1 + \theta_2 = \frac{1}{2}.$$

Invoking (23) and solving for the optimal values of β_i yields β_i^{RS} in (22a). This completes the proof. □

Corollary 3: In the general case of $n > 2$, the optimal risk-neutral bidding strategy is

$$\alpha_i^* = \frac{1}{n+1}\eta_i^{-1}, \tag{24}$$

and the total profit Π in (18) is optimized with

$$g^{RN}(\boldsymbol{\alpha}^*) = \frac{n}{(n+1)^2}(x_r^o)^2.$$

Proof: When $\beta_i = 0$, the matrix equation (11) reduces to

$$\left[\text{diag}\{\eta_1, \cdots, \eta_n\} + \mathbf{1}\boldsymbol{\eta}^T\right]\boldsymbol{\alpha} = \mathbf{1},$$

where $\boldsymbol{\eta} = [\eta_1 \ \cdots \ \eta_n]^T$ and $\mathbf{1} = [1 \ \cdots \ 1]^T$. It is simple to verify that the solution is given by (24). And, the corresponding total profit can be calculated in terms of (12) and (13). □

Corollary 3 provides a closed-form Nash solution to the risk-neutral bidding problem, and the solution implicitly assumes that overbidding (that is, $\alpha_i^* > 1$ for some i) is allowed. Physically, operator i is overbidding if it commits to deliver more power than its forecast capacity x_i^o into the day-ahead market, exposing itself to shortfall real-time. It follows that Nash strategy (24) renders no overbidding if and only if

$$\eta_i \geq \frac{1}{n+1} \quad \forall i.$$

Since $\sum_i \eta_i = 1$, we know that

$$\min_i \eta_i \leq \frac{1}{n}.$$

Therefore, Nash strategy (24) does not produce overbidding if

$$\min_i \eta_i \in \left[\frac{1}{n+1},\, \frac{1}{n}\right], \quad \max_i \eta_i \leq \frac{2}{n+1}, \tag{25}$$

which implies that the maximum capacity difference among the renewable operators is limited to $1/(n+1)$. Should overbidding be forbidden while condition (25) fails, the bidding strategy is given by the following corollary instead of that in corollary 3. In corollary 4, set Ω_{RN} prescribes the set of renewable operators whose bids are saturated, and the set is defined through recursion (28) and with initial set value (27), in which the bid-saturating threshold on η_i is increased at each step until $\eta_i \geq \frac{\sum_{l\notin\Omega_{\rm RN}} \eta_l}{n+1-|\Omega_{\rm RN}|}$ holds for all $i \notin \Omega_{RN}$. This ensures that $\alpha_i^* \leq 1$ for all i.

Corollary 4: In the case that overbidding is prohibited, the optimal risk-neutral bidding strategy is

$$\alpha_i^* = \begin{cases} \dfrac{\sum_{l\notin\Omega_{\rm RN}} \eta_l}{n+1-|\Omega_{\rm RN}|}\,\eta_i^{-1} & i \notin \Omega_{\rm RN}, \\ 1 & i \in \Omega_{\rm RN}, \end{cases} \tag{26}$$

where $\Omega_{\rm RN}$ is the bid-saturated set defined by $\Omega_{\rm RN}^{(n)}$, with initial value

$$\Omega_{\rm RN}^{(1)} = \left\{ j \in \Omega : \eta_j < \tfrac{1}{n+1} \right\}, \tag{27}$$

and the kth recursion

$$\Omega_{\rm RN}^{(k+1)} = \Omega_{\rm RN}^{(k)} \cup \left\{ j \notin \Omega_{\rm RN}^{(k)} : \eta_j < \frac{\sum_{l\notin\Omega_{\rm RN}^{(k)}} \eta_l}{n+1-|\Omega_{\rm RN}^{(k)}|} \right\}. \tag{28}$$

Furthermore, the total profit Π in (18) is optimized with

$$g^{RN}(\boldsymbol{\alpha}^*) = \frac{\sum_{j\notin\Omega_{\rm RN}} \eta_j}{(n+1-|\Omega_{\rm RN}|)^2} \left[n - |\Omega_{\rm RN}| + \sum_{j\in\Omega_{\rm RN}} \eta_j \right] (x_r^o)^2.$$

Proof: It follows from (17) with $\beta_i = 0$ and from $\sum_i \eta_i = 1$ that, for all i,

$$\sum_{j\in\Omega} \alpha_j \eta_j + \alpha_i \eta_i = 1.$$

Since $\alpha_j = 1$ for $j \in \Omega_{\rm RN}$, the above equation becomes

$$\sum_{j\notin\Omega_{\rm RN}} \alpha_j \eta_j + \alpha_i \eta_i = \sum_{l\notin\Omega_{\rm RN}} \eta_l, \qquad \forall\, i \notin \Omega_{\rm RN},$$

which implies $\alpha_i \eta_i$ is invariant for $i \notin \Omega_{\rm RN}$. Summing up the above equation for all $i \notin \Omega_{\rm RN}$, we have

$$\sum_{j\notin\Omega_{\rm RN}} \alpha_j \eta_j = \frac{n - |\Omega_{\rm RN}|}{n+1-|\Omega_{\rm RN}|} \sum_{l\notin\Omega_{\rm RN}} \eta_l.$$

Combining the above two equations yield solution (26).

It follows from (13) that

$$\begin{aligned} f(\boldsymbol{\alpha}^*) &= \sum_{i\notin\Omega_{\rm RN}} \alpha_i \eta_i + \sum_{j\in\Omega_{\rm RN}} \eta_j \\ &= \frac{n - |\Omega_{\rm RN}|}{n+1-|\Omega_{\rm RN}|} + \frac{\sum_{j\in\Omega_{\rm RN}} \eta_j}{n+1-|\Omega_{\rm RN}|}, \end{aligned}$$

and $g(\boldsymbol{\alpha}^*) = f(\boldsymbol{\alpha}^*)\big[1 - f(\boldsymbol{\alpha}^*)\big](x_r^o)^2$, which gives the result in the statement. □

The significant outcome of theorem 1 and its corollaries is that, while the total profit under noncooperative risk neutral Nash equilibrium is typically worse than that of cooperative coalition solution, risk aware Nash strategies could be employed to recover the same profit as the cooperative coalition's. Indeed, corollary 2 provides an infinite number of combinations in risk tolerance levels as long as the aggregate risk tolerance of $(\beta_1 + \beta_2)$ assumes an optimal risk seeking value. In the next section, this result is extended to a centralized solution to the general case of n market participants.

IV. Risk Aware Nash Solutions with Pareto Payoff

The results in Section III establish that, for any fixed choice of the risk parameters $\{\beta_i\}_{i=1}^n$, the noncooperative bidding game among renewable generators admits a unique Nash equilibrium in terms of the bidding coefficients $\{\alpha_i^*\}_{i=1}^n$. However, the resulting Nash equilibrium is generally not Pareto optimal with respect to the total expected profit, and there are an infinite number choices of risk tolerance level choices that yield the same Pareto solution. In this section, we show that the Nash equilibrium can be systematically shifted to a Pareto-optimal operating point by appropriately selecting the risk parameters $\{\beta_i\}$. Importantly, this Pareto enhancement does not require coordination on the bidding variables α_i themselves, but only on the aggregate risk exposure across agents.

Theorem 2: There exists a family of risk tolerance levels in $\boldsymbol{\beta} = [\beta_1 \ \cdots \ \beta_n]^T$ such that the corresponding Nash equilibrium $\boldsymbol{\alpha}^* = [\alpha_1^* \ \cdots \ \alpha_n^*]^T$ satisfying (11) achieves the maximum aggregate profit:

$$g^{RS}(\boldsymbol{\alpha}^*) = \frac{1}{4}(x_r^o)^2 \triangleq g^{\max},$$

where $\boldsymbol{\alpha}^*(\boldsymbol{\beta})$ denote the corresponding Nash equilibrium strategies. Specifically, any admissible set of risk tolerance levels satisfying

$$f(\boldsymbol{\alpha}^*(\boldsymbol{\beta})) = \frac{1}{2}, \tag{29}$$

will result in a Pareto-optimal Nash equilibrium. Furthermore, the Pareto optimal total profit is reached if and only if the aggregate risk tolerance satisfies the following condition:

$$\sum_{i\in\Omega}\beta_i=\frac{1-n}{4a_2}, \tag{30}$$

and the corresponding bidding strategy is

$$\alpha_i^*=\frac{1}{2}(1+4a_2\beta_i)\eta_i^{-1}. \tag{31}$$

Proof: Define the mapping $\Phi:\mathbb{R}^n\to\mathbb{R}$ as

$$\Phi(\boldsymbol{\beta})=f(\boldsymbol{\alpha}^*(\boldsymbol{\beta})),$$

where $\boldsymbol{\alpha}^*(\boldsymbol{\beta})$ denotes the Nash equilibrium solution of (11). It follows from (11) that $\Phi(\cdot)$ is continuous. To establish existence, we show that $\Phi(\cdot)$ attains values below and above $\frac{1}{2}$:

- It follows from (17) that, as $\beta_i\to-\frac{1}{4a_2}$ for all i, $\alpha_i^*\to 0$ and hence $\Phi(\boldsymbol{\beta})\to 0<\frac{1}{2}$.
- It follows from corollary 3 that, when $\beta_i=0$ for all i, $\alpha_i^*=\frac{1}{n+1}\eta_i^{-1}$ and consequently $\Phi(\mathbf{0})=f(\boldsymbol{\alpha}^*(\mathbf{0}))=\frac{n}{n+1}>\frac{1}{2}$ for $n\geq 2$.

Therefore, by continuity of $\Phi(\cdot)$ and the intermediate value theorem [19], there exists $\boldsymbol{\beta}^*$ such that $\Phi(\boldsymbol{\beta}^*)=\frac{1}{2}$, achieving $g=g^{\max}$.

Dividing both sides of (17) by x_r^o and using $\eta_i=x_i^o/x_r^o$ and $\sum_j(1-\alpha_j^*)x_j^o/x_r^o=1-\sum_j\alpha_j^*\eta_j=1-f(\boldsymbol{\alpha}^*)$, it follows that, at the Nash equilibrium,

$$\alpha_i^*\eta_i=(1+4a_2\beta_i)\big(1-f(\boldsymbol{\alpha}^*)\big). \tag{32}$$

Summing both sides over all i yields

$$f(\boldsymbol{\alpha}^*)=\big(1-f(\boldsymbol{\alpha}^*)\big)\left(n+4a_2\sum_{i\in\Omega}\beta_i\right),$$

which renders the following result:

$$f(\boldsymbol{\alpha}^*)=\frac{n+4a_2\sum_{i\in\Omega}\beta_i}{1+n+4a_2\sum_{i\in\Omega}\beta_i}.$$

Therefore, we know that the Pareto optimality condition (29) holds if and only if

$$n+4a_2\sum_{i\in\Omega}\beta_i=1,$$

which yields (30). Substituting the results in (30) and (29) into (32) yields the bidding strategy (31). □

The Pareto optimal condition (29) is a constraint only on the aggregate risk tolerance level; therefore, infinitely many heterogeneous risk allocations $\{\beta_i\}$ yield Pareto-optimal Nash equilibria. If there is only one renewable generator ($n=1$ or one coalition of all the operators), condition (30) reduces to $\beta_1=0$, which confirms, as in corollary 1, that, in the absence of competition, risk-seeking or risk-averse behavior provides no strategic advantage. In the case of duopoly, condition (30) reduces to (22a) in corollary 2. As the number of renewable operators increases, the aggregate risk tolerance $\sum_{i\in\Omega}\beta_i$ decreases monotonically. In the limit of $n\to\infty$, it is better to restate condition (30) on a per-agent basis, that is,

$$\frac{1}{n}\sum_{i\in\Omega}\beta_i=\frac{1-n}{4a_2n}=-\frac{1}{4a_2}+\frac{1}{4a_2n}\to-\frac{1}{4a_2}\ \text{as}\ n\to\infty,$$

which coincides with the lower bound on β_i as stated in theorem 1. This means that, as the number of renewable operators grows, Pareto optimality forces the average risk tolerance level to approach its feasibility lower bound from above. In the large-population limit, aggregate optimal performance is therefore achieved when the collective risk tolerance reaches its lower bound. This reveals an emergent mean-field structure: rather than requiring coordination of individual bidding strategies, Pareto efficiency is recovered through convergence of the aggregate risk profile to its lower limit. This connection is reminiscent of mean-field game theory [20], [21], in which the strategic interactions of a large population of agents can often be characterized through a single aggregate quantity. Here the average risk tolerance $\frac{1}{n}\sum_i\beta_i$ plays the role of the mean-field state, and Pareto efficiency emerges as this quantity converges to a well-defined limit, without requiring direct coordination of individual bids.

All the renewable entities may adopt the same risk tolerance level as

$$\beta_i^*=\beta_0,\quad\forall i\in\Omega. \tag{33}$$

Note that all the entities employing a common risk seeking level is a fair policy enforceable by the utility and is also supported by the Nash equilibria formed by all the operators. In this special case, theorem 2 and corollary 4 reduce to the following corollary. It should be noted that, when implementing equal risk seeking strategy (35) or (36), the renewable operators do not need to know the value of parameter a_2 even though an explicit calculation of the equal risk tolerance level in (33) would require the value of a_2.

Corollary 5: The equal risk seeking (ERS) bidding strategy (under (33)) is given as follows:

(a) If set

$$\Omega_{\text{ERS}}^0=\left\{j\in\Omega:\eta_j<\tfrac{1}{2n}\right\} \tag{34}$$

is empty, then

$$\alpha_i^*=\frac{1}{2n}\eta_i^{-1}. \tag{35}$$

(b) When the set Ω_{ERS}^0 is non-empty, strategy (35) would yield overbidding (that is, $\alpha_i^*>1$) for some of the operators, and the constrained ERS and Pareto-optimal bidding strategy is

$$\alpha_i^*=\begin{cases}1, & i\in\Omega_{\text{ERS}},\\[4pt] \dfrac{\sum_{j\notin\Omega_{\text{ERS}}}\eta_j-\sum_{l\in\Omega_{\text{ERS}}}\eta_l}{2\,(n-|\Omega_{\text{ERS}}|)\,\eta_i}, & i\notin\Omega_{\text{ERS}},\end{cases} \tag{36}$$

where

$$\Omega_{\mathrm{ERS}} \triangleq \left\{ i \in \Omega : \quad \eta_i < \tfrac{1}{2}\big(\max_{l\in\Omega} \eta_l + \min_{j\in\Omega} \eta_j\big) \right\}. \qquad (37)$$

Under strategy (35) or (36), $\alpha_i^* \le 1$ for all i, and the total profit Π in (18) is maximized with $f(\boldsymbol{\alpha}^*) = \frac{1}{2}$ and $g^{ERS}(\boldsymbol{\alpha}^*) = \frac{1}{4}(x_r^o)^2 = g^{\max}$.

Proof: Case (a) follows from Ω^0_{ERS} being empty that, by theorem 2, condition (30) can be met by choosing (33) with

$$\beta_0 = \tfrac{1-n}{4a_2 n}. \qquad (38)$$

Substituting the above into (31) yields strategy (35). It follows from (12) that, under strategy (35), $f(\boldsymbol{\alpha}^*) = \frac{1}{2n}\sum_i 1 = \frac{1}{2}$, and $g^{ERS}(\boldsymbol{\alpha}^*) = \frac{1}{4}(x_r^o)^2 = g^{\max}$.

In case (b), β_i^* can no longer be chosen according to (33). Instead, β_i^* is chosen to be equal within each of the two subgroups and as

$$\beta_i^* \;=\; \begin{cases} \frac{2\eta_i - 1}{4a_2} & i \in \Omega_{\mathrm{ERS}}, \\ \dfrac{\sum_{j\notin\Omega_{\mathrm{ERS}}} \eta_j - \sum_{l\in\Omega_{\mathrm{ERS}}} \eta_j}{4a_2(n - |\Omega_{\mathrm{ERS}}|)} - \dfrac{1}{4a_2}, & i \notin \Omega_{\mathrm{ERS}} \end{cases}. \qquad (39)$$

It follows that

$$\begin{aligned} &\sum_i 4a_2\beta_i^* \\ &= \sum_{i\in\Omega_{\mathrm{ERS}}} (2\eta_i - 1) + \sum_{i\notin\Omega_{\mathrm{ERS}}} \left(\tfrac{\sum_{j\notin\Omega_{\mathrm{ERS}}} \eta_j - \sum_{l\in\Omega_{\mathrm{ERS}}} \eta_j}{n - |\Omega_{\mathrm{ERS}}|} - 1 \right) \\ &= 1 - n, \end{aligned}$$

which meets condition (30) in theorem 2. We also know from (32) in the proof of theorem 2 that, in order to achieve Pareto optimal solution of $f(\boldsymbol{\alpha}^*) = \frac{1}{2}$ and $g^{ERS}(\boldsymbol{\alpha}^*) = \frac{1}{4}(x_r^o)^2 = g^{\max}$, condition

$$\eta_i\alpha_i^* = \tfrac{1}{2}(1 + 4a_2\beta_i^*)$$

has to hold for every operator. Substituting (39) into the above yields optimal bidding strategy (36).

To establish $\alpha_i^* \le 1$ for all $i \in \Omega$, it suffices to show that, for $i \notin \Omega_{\mathrm{ERS}}$,

$$\sum_{j\notin\Omega_{\mathrm{ERS}}} \eta_j - \sum_{l\in\Omega_{\mathrm{ERS}}} \eta_j \le 2(n - |\Omega_{\mathrm{ERS}}|) \min_{i\notin\Omega_{\mathrm{ERS}}} \eta_i. \qquad (40)$$

It follows from the definition of Ω_{ERS} that

$$\begin{aligned} &\sum_{j\notin\Omega_{\mathrm{ERS}}} \eta_j - \sum_{l\in\Omega_{\mathrm{ERS}}} \eta_j \\ \le\;& (n - |\Omega_{\mathrm{ERS}}|) \max_{j\notin\Omega_{\mathrm{ERS}}} \eta_j - |\Omega_{\mathrm{ERS}}| \min_{l\in\Omega_{\mathrm{ERS}}} \eta_l \\ =\;& (n - |\Omega_{\mathrm{ERS}}|) \left(\max_{j\notin\Omega_{\mathrm{ERS}}} \eta_j + \min_{l\in\Omega_{\mathrm{ERS}}} \eta_l \right) - n \min_{l\in\Omega_{\mathrm{ERS}}} \eta_l \\ =\;& (n - |\Omega_{\mathrm{ERS}}|) \left(\max_{j\notin\Omega} \eta_j + \min_{l\in\Omega} \eta_l \right) - n \min_{l\in\Omega_{\mathrm{ERS}}} \eta_l \\ \le\;& 2(n - |\Omega_{\mathrm{ERS}}|) \left[\tfrac{1}{2} \left(\max_{j\notin\Omega} \eta_j + \min_{l\in\Omega} \eta_l \right) \right] \\ \le\;& 2(n - |\Omega_{\mathrm{ERS}}|) \min_{i\notin\Omega_{\mathrm{ERS}}} \eta_i, \end{aligned}$$

which validates inequality (40). This concludes the overall proof. □

When there is no restriction on overbidding, every operator prefers the Pareto-optimal (risk seeking) Nash equilibrium to the risk-neutral Nash equilibrium. To see this conclusion, note from (14) that operator i's mean profit is given by

$$\Pi_i = 2a_2\, \alpha_i (1 - f)(x_r^o)^2 \eta_i + C_i,$$

where C_i is the sum of those terms that are in (14) and independent of bidding strategies used. It follows from corollary 5(a) and corollary 3 that, under the ERS strategy,

$$\alpha_i^*(1 - f^{ERS}) = \tfrac{1}{2n}\eta_i^{-1} \cdot \tfrac{1}{2} = \tfrac{1}{4n}\eta_i^{-1};$$

and under the RN strategy,

$$\alpha_i^{RN}(1 - f^{RN}) = \tfrac{1}{n+1}\eta_i^{-1} \cdot \tfrac{1}{n+1} = \tfrac{1}{(n+1)^2}\eta_i^{-1}.$$

Hence, we have

$$\Pi_i^{RS} - \Pi_i^{RN} = \frac{a_2(n-1)^2}{2n(n+1)^2}(x_r^o)^2 > 0 \quad \forall\, n \ge 2.$$

All operators gain equally from adopting the risk seeking strategy and moving from RN Nash to the Pareto-optimal equilibrium, confirming that the proposed bidding mechanism is individually rational.

When overbidding is not permitted, each of the operators can individually check for itself $\eta_i \ge \frac{1}{2}$ as the condition of whether an overbidding would occur. It should be noted that this condition without any overbidding in the ERS setup is much more relaxed than (25) in the RN setup. If Ω^0_{ERS} in (34) is empty, there is no overbidding, and all the operators can adopt bidding strategy (35). If Ω^0_{ERS} in (34) is not empty, at least one of the operators will overbid under strategy (35). To ensure no overbidding for all the operators while achieving pareto-optimal payoffs, those operators corresponding to set Ω_{ERS} in (37) are forced to saturate their bids as $\alpha_i^* = 1$, which makes their risk seeking levels individually different for distinct values of η_i. The rest of the operators not contained in set Ω_{ERS} share a common risk seeking level, and their bids are unsaturated. These results are mathematically shown in (36) and (39). Note also that, unlike the recursive process of determining set set Ω_{RN} in corollary 4, set Ω_{ERS} in corollary 5 is determined in one step (but somewhat conservatively).

In what follows, the Pareto-optimal bidding strategy is cast into a distributed optimization algorithm so that minimum information sharing is needed and privacy is protected.

V. Distributed Risk Aware Solutions

It follows from (18), theorem 1, theorem 2, and corollary 5 that the Pareto optimal total profit

$$\begin{aligned}\Pi^{\max} &= 2a_2\left[x^o_{n+1}-\frac{3}{4}x^o_r\right]x^o_r+a_1x^o_r\\ &\quad +2a_2\left[\sum_{i\in\Omega}\sigma_{iL}-\sum_{i\in\Omega}\sum_{j\in\Omega}\sigma_{ij}\right],\end{aligned}$$

is reached provided that the renewable operators use bidding strategy (35). Direct implementation of strategy (35) by the ith operator requires the following information:

- The total of renewable operators participating in the markets, n, which could be either fixed or changing;
- The percentage of renewable generation contributed by the ith operator with respect to the total renewable generation, η_i.

In addition, implementation of strategy (36) requires the following extra information:

- Maximum and minimum of η_i;
- Set Ω_{ERS}, its cardinality $|\Omega_{\text{ERS}}|$, as well as the sums of $\sum_{j\notin\Omega_{\text{ERS}}}\eta_j$ and $\sum_{l\in\Omega_{\text{ERS}}}\eta_j$.

In what follows, all the above information are determined distributively and iteratively, based on the following lemma [22].

Lemma 2: Consider n agents and their average/maximum/minimum consensus protocols:

$$\begin{aligned}z_i(k+1) &= z_i(k)+\epsilon_i\sum_{j\in\mathcal{N}_i}[z_j(k)-z_i(k)],\\ \overline{z}_i(k+1) &= \max_{j\in\{i\}\cup\mathcal{N}_i}\overline{z}_j(k),\\ \underline{z}_i(k+1) &= \min_{j\in\{i\}\cup\mathcal{N}_i}\underline{z}_j(k),\end{aligned}$$

where $\mathcal{N}_i$ denotes the set of neighbors of the ith agent, and $\epsilon_i>0$ is the consensus gain satisfying $\epsilon_i<\frac{1}{\max_i|\mathcal{N}_i|}$. Then, as long as the communication topology is connected, the average/maximum/minimum consensus is reached in the sense that

$$\begin{aligned}\lim_{k\to\infty}z_i(k) &= \frac{1}{m}\sum_{j=1}^{m}z_j(0),\quad \forall i;\\ \lim_{k\to n}\overline{z}_i(k) &= \max_{i\in\Omega}\overline{z}_j(0),\\ \lim_{k\to n}\underline{z}_i(k) &= \min_{i\in\Omega}\underline{z}_j(0).\end{aligned}$$

The proposed distributed algorithm is given by Algorithm 1. Since every participating renewable operator must communicate with the utility, the communication topology among all the participating operators and the utility is guaranteed to be connected. It follows from Lemma 2, initial conditions $\xi_i(0)$ and $\psi_i(0)$, and protocols (41) and (42) that $\xi_i(k)\to\frac{1}{n+1}$ and $\psi_i(k)\to\frac{1}{n+1}x^o_r$. Therefore, the estimate of n, η_i and β'_i can be computed distributively. There is no explicit sharing of private information, and required information is generated through diffusion of the consensus protocols (41) and (42).

Algorithm 1 Distributed Risk-Aware Nash Optimization

Require: $c>0$; $\xi_{n+1}(0)=1$, $\psi_{n+1}(0)=0$; $\xi_i(0)=0$, $\psi_i(0)=x^o_i$

Ensure: α^*_i for each operator i

Phase 1. Run average consensus (41)–(42) until $|\xi_i(k)-\xi_i(k-1)|\le c$:

$$\xi_i(k+1)=\xi_i(k)+\epsilon_i\sum_{j\in\mathcal{N}_i}[\xi_j(k)-\xi_i(k)],\tag{41}$$

$$\psi_i(k+1)=\psi_i(k)+\epsilon_i\sum_{j\in\mathcal{N}_i}[\psi_j(k)-\psi_i(k)].\tag{42}$$

Phase 2. Compute individually $\hat{n}_i=1/\xi_i-1$, $\eta_i=x^o_i\xi_i/\psi_i$, and $\beta'_i\triangleq a_2\beta_i=(1-\hat{n}_i)/(4\hat{n}_i)$

Phase 3. Compute max/min consensus $\eta_{\max}=\max_{i\in\Omega}\eta_i$, $\eta_{\min}=\min_{i\in\Omega}\eta_i$, and $\eta_c=(\eta_{\max}+\eta_{\min})/2$.

if $\eta_{\min}\ge 1/(2\hat{n}_i)$ **then**

  $\alpha^*_i=1/(2\hat{n}_i\eta_i)$

else

  $i\in\Omega_{\text{ERS}}$ if $\eta_i<\eta_c$; otherwise $i\notin\Omega_{\text{ERS}}$

  Phase 4. For $i\notin\Omega_{\text{ERS}}$, use a version of (41) to calculate cardinality $(n-|\Omega_{\text{ERS}}|)$, and use a version of (42) to compute $\sum_{j\notin\Omega_{\text{ERS}}}\eta_j$.

  if $i\in\Omega_{\text{ERS}}$ **then**

    $\alpha^*_i=1$

  else

    $\alpha^*_i=\dfrac{2\left(\sum_{j\notin\Omega_{\text{ERS}}}\eta_j\right)-1}{2\left(n-|\Omega_{\text{ERS}}|\right)\eta_i}$

  end if

end if

Algorithm 1 operates over an undirected, static communication graph in which every renewable operator maintains a direct link with the utility, forming a star topology $K_{1,n}$. This structure guarantees connectivity for any $n\ge 1$: since every operator communicates with the utility, the graph is connected by construction and requires no additional peer-to-peer links.

The consensus protocols (41)–(42) converge geometrically at a rate determined by the algebraic connectivity $\lambda_2(L)$, the second-smallest eigenvalue of the graph Laplacian L. By Theorem 3.3 of [22], the convergence error satisfies $\|z(k)-z^\infty\mathbf{1}\|\le(1-\epsilon\lambda_2)^k\|z(0)-z^\infty\mathbf{1}\|$, so the number of iterations to reach ε-accuracy scales as $O(\log(1/\varepsilon)/\lambda_2)$. For the star graph $K_{1,n}$, the Laplacian eigenvalues are 0, 1 (with multiplicity $n-1$), and $n+1$ (with multiplicity 1), giving $\lambda_2=1$ independently of n. Consequently, Algorithm 1 converges at a rate that does not degrade as the number of operators increases, making it well-suited for large-scale markets. With step size $\epsilon<1/n$, the per-iteration contraction factor is $\rho=1-\epsilon<1$. The same convergence guarantee applies to the auxiliary σ and μ consensus rounds in the constrained branch, both of which run over the same star graph at the same geometric rate. The max-/min-consensus on η_i terminates in a single round of forward-and-back communication through the utility hub on $K_{1,n}$.

VI. Illustrative Example

To demonstrate the proposed analytical and algorithmic results, a three-operator case study is used in which both the centralized solution and the distributed solution are computed and compared across multiple scenarios.

TABLE I
Example Parameters

Parameter	n	a_2	a_1	a_0	x_1^o	x_2^o	x_3^o	x_L^o
Value	3	0.1	1	0	120	100	80	400

A. Basic Setup

Consider the day-ahead/real-time electricity markets with three renewable generation operators $i \in \{1,2,3\}$ and one conventional generation aggregator (as the slack bus). The power system parameters are provided in Table I. It follows that the day-ahead forecasts for the renewables are x_1^o, x_2^o, x_3^o in MW, respectively, that $x_4^o = P_L^o$ is the load forecast, and that the objective function of traditional generation is

$$J(P_g) = 0.1P_g^2 + P_g, \quad P_g = x_4 - (x_1 + x_2 + x_3).$$

Real-time realizations are $x_i \sim \mathcal{N}(x_i^o, \sigma_i^2)$, with standard deviations of $\sigma_1 = 12\,\text{MW}$, $\sigma_2 = 10\,\text{MW}$, $\sigma_3 = 8\,\text{MW}$, load uncertainty is $\sigma_L = 20\,\text{MW}$, cross-correlations are $\rho_{ij} = 0.3$ among renewable operators, and $\rho_{iL} = 0.2$ among renewables and the load. As seen from (18), the stochastic terms enter total average profit Π only as the constant offset $\Delta\Pi$ in (19), which is identical across all strategies and does not affect the optimization results of choosing α_i or β_i. Hence, all optimal bidding strategies and the strategic comparisons reported below are independent of the uncertainty parameters; the profit values in Table II are those excluding the constant offset.

B. Analytical and Centralized Nash Solutions

Renewable operators have the following choices:

- Coalition bidding strategy (8) (if all agree to join);
- Risk seeking Pareto-optimal Nash strategy (11), that is,

$$\alpha_1^* = \frac{(1 + 4a_2\beta_1)\eta_1^{-1}}{4\,(1 + a_2(\beta_1 + \beta_2 + \beta_3))}, \tag{43a}$$

$$\alpha_2^* = \frac{(1 + 4a_2\beta_2)\eta_2^{-1}}{4\,(1 + a_2(\beta_1 + \beta_2 + \beta_3))}, \tag{43b}$$

$$\alpha_3^* = \frac{(1 + 4a_2\beta_3)\eta_3^{-1}}{4\,(1 + a_2(\beta_1 + \beta_2 + \beta_3))}. \tag{43c}$$

It follows from Lemma 1 that, if all operators cooperate and adopt the coalition strategy, their bids are $\alpha^* = \frac{1}{2}$ (with $\beta^* = 0$), which yields $f(\boldsymbol{\alpha}^*) = \frac{1}{2}$, $g^C(\boldsymbol{\alpha}^*) = 22{,}500\,\text{MW}^2$, and the total profit of $\Pi^C = 10{,}800\,\$/\text{hr}$. This serves as the upper bound for the collective profit.

It follows from (43) with $\beta_i = 0$ that the risk neutral Nash bidding strategies are $\boldsymbol{\alpha}^* = (\alpha_1^*, \alpha_2^*, \alpha_3^*) = (0.625, 0.75, 0.9375)$, which results in $f(\boldsymbol{\alpha}^*) = 0.75$, $g(\boldsymbol{\alpha}^*) = 16{,}875\,\text{MW}^2$, and the total profit $\Pi^{RN} = 9{,}675\,\$/\text{hr}$. In this case, the collective profit is 10.42% below that under the coalition solution. On the other hand, risk seeking Pareto-optimal Nash bidding strategies in (43) allow different choices of risk seeking levels β_i by different operators as long as constraint (30) is satisfied. Admissible choices of β_i form in Figure 2. Should all renewable operators agree to choose the same risk tolerance level, we know from (30) that $\beta_i^* = -\frac{5}{3}$ (equivalently $\beta_i'^* = a_2\beta_i^* = -\frac{1}{6}$), which yields $\boldsymbol{\alpha}^* = (0.4167, 0.5000, 0.6250)$, resulting in $f(\boldsymbol{\alpha}^*) = 0.5$, $g(\boldsymbol{\alpha}^*) = 22{,}500\,\text{MW}^2$, and the total profit $\Pi^{RS} = 10{,}800\,\$/\text{hr}$. This risk seeking Pareto-optimal Nash solution fully recovers the total profit under the coalition strategy.

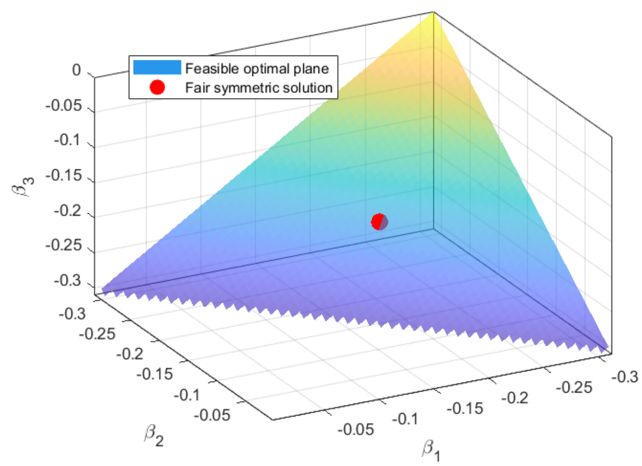

Fig. 2. Feasible risk-seeking Pareto allocations for the example ($n = 3$).

Table II and Figures 3–4 summarize the performance of all four strategies. The risk-neutral Nash equilibrium incurs a 10.42% profit loss relative to the coalition, while the risk-seeking Pareto-optimal Nash equilibrium fully recovers the coalition-level profit through risk coordination alone, without any change in the game structure. The distributed Algorithm 1 exactly reproduces the centralized Pareto-optimal result: each operator independently estimates $\hat{n}_i = 3$ and $\hat{x}_r^o = 300\,\text{MW}$ from the consensus protocols, computes $\beta_i' = -\frac{1}{6}$, and submits the same bids and achieves the same profit as the centralized symmetric Pareto case.

Figure 3 shows that the risk-neutral Nash bids are noticeably more aggressive — particularly for Op 3, whose small capacity share ($\eta_3 = 0.267$) drives α_3^* close to 1. Risk-seeking Pareto coordination uniformly lowers and

TABLE II
Performance Comparison of Bidding Strategies ($n = 3$)

Strategy	$\boldsymbol{\alpha}^*$	f	$g\,(\mathrm{MW}^2)$	$\sum_i \Pi_i$ (\$/hr)
Coalition	(0.50, 0.50, 0.50)	0.50	22,500	10,800
RN Nash	(0.63, 0.75, 0.94)	0.75	16,875	9,675
RS Pareto Nash	(0.42, 0.50, 0.63)	0.50	22,500	10,800
Algo. 1 (Dist.)	(0.42, 0.50, 0.63)	0.50	22,500	10,800

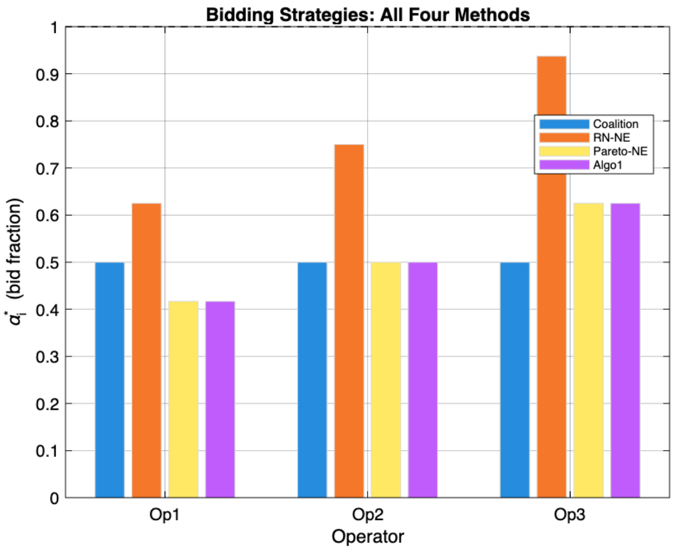

Fig. 3. Bidding strategies α_i^* for all four strategies.

rebalances the bids toward the coalition optimum, with Algorithm 1 exactly reproducing the centralized result.

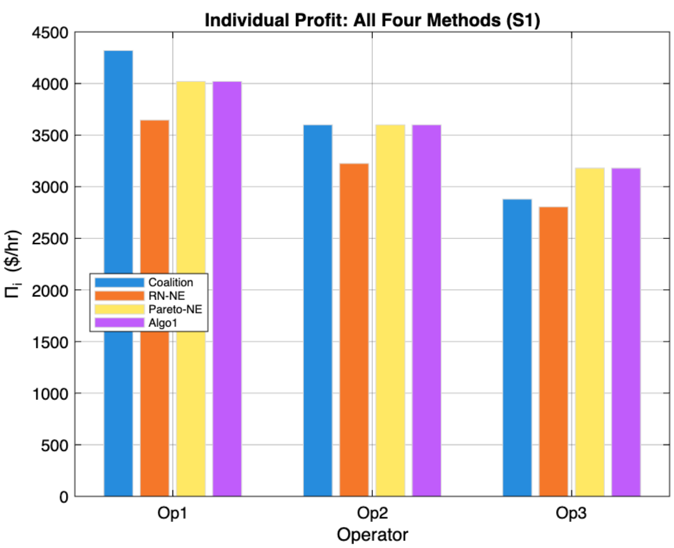

Fig. 4. Individual profits Π_i for all four strategies (Scenario 1).

Figure 4 confirms that the profit improvement under Pareto coordination is shared across all operators: each Π_i rises relative to the risk-neutral Nash case, so the move from RN to RS is a Pareto improvement in the game-theoretic sense — no operator is made worse off by adopting the risk-seeking strategy. Figure 5 further shows the per-unit perspective: under risk-neutral Nash, the smaller Op 3 earns less per MW of installed capacity, while Pareto coordination through risk-seeking restores equal per-unit returns.

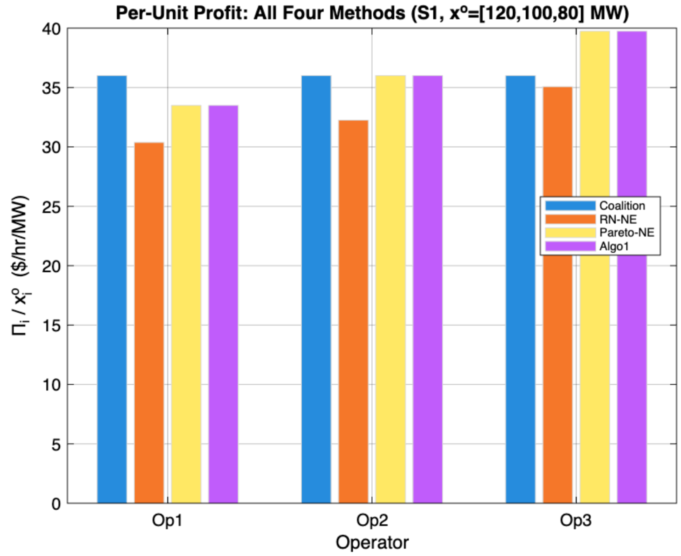

Fig. 5. Per-unit profit Π_i/x_i^o (\$/hr/MW) for all four strategies (Scenario 1). Normalizing by capacity reveals that smaller operators earn disproportionately less per MW under risk-neutral Nash; Pareto coordination through risk-seeking equalizes the per-unit returns across operators.

C. Heterogeneous Pareto Risk Allocations

Theorem 2 states that any $\boldsymbol{\beta}$ satisfying $\sum_i \beta_i' = (1-n)/4$ achieves Pareto optimality, yielding infinitely many valid heterogeneous allocations. Three representative allocations for the base parameters illustrate this flexibility:

- Symmetric ERS: $\beta_i' = -\frac{1}{6}$ for all i (Corollary 5a). Bids are $\boldsymbol{\alpha}^* = (0.417, 0.500, 0.625)$, $f = 0.5$, $g = g^{\max}$.
- Hetero A: $\boldsymbol{\beta}' = (-0.10, -0.17, -0.23)$ — Op 3 (smallest share) is most risk-seeking. Bids become $\boldsymbol{\alpha}^* = (0.750, 0.480, 0.150)$, with $f = 0.5$ and $g = g^{\max}$ fully preserved.
- Hetero B: $\boldsymbol{\beta}' = (-0.23, -0.17, -0.10)$ — Op 1 is most risk-seeking. Although $\sum_i \beta_i' = -0.5$ satisfies the Pareto aggregate condition, Op 3's low risk tolerance yields $\alpha_3^* \approx 1.12 > 1$, violating the capacity constraint. Clamping reduces f below $\frac{1}{2}$ and incurs an efficiency loss.

Symmetric ERS and Hetero A both achieve $\Pi^{\text{total}} = 10{,}800\,\$/\text{hr}$, but distribute profits differently: Hetero A shifts profit toward Op 1 (higher bid) and away from Op 3 (lower bid). This gives a regulator or cooperative agreement a mechanism to adjust equity without sacrificing aggregate efficiency — a direct consequence of the infinite-solution structure of Theorem 2.

D. Overbidding Impact

From (34), overbidding occurs for operator i at the ERS optimum whenever $\eta_i < \frac{1}{2n}$. For the original parameters in

Table I, $\eta_i \in \{0.40, 0.33, 0.27\}$ all exceed $\frac{1}{6}$, so $\Omega_{\mathrm{ERS}} = \varnothing$ and no overbidding occurs; Scenario 1 in Table III confirms that the symmetric Pareto allocation $\beta'^* = -\frac{1}{6}$ achieves $f = 0.5$ and $g = g^{\max}$.

To illustrate the heterogeneous-β branch of Corollary 5b, we introduce Scenario 2 by setting $x_3^o = 30\,\mathrm{MW}$, giving $x_r^o = 250\,\mathrm{MW}$ and $\boldsymbol{\eta} = (0.48, 0.40, 0.12)$. Operator 3 has $\eta_3 < \frac{1}{2n} = \frac{1}{6}$, so $\Omega_{\mathrm{ERS}} = \{3\}$ and the homogeneous ERS bid is $\alpha_3^* = \frac{1}{2n\eta_3} \approx 1.39 > 1$.

The partition (37) gives $\eta_{\max} = 0.48$, $\eta_{\min} = 0.12$, midpoint $\eta_c = 0.30$, hence $\Omega_{\mathrm{ERS}} = \{3\}$ and $\Omega \setminus \Omega_{\mathrm{ERS}} = \{1, 2\}$. Substituting into (39)–(36) with $\sum_{j \notin \Omega_{\mathrm{ERS}}} \eta_j = 0.88$, $\sum_{j \in \Omega_{\mathrm{ERS}}} \eta_j = 0.12$, $n - |\Omega_{\mathrm{ERS}}| = 2$:

$$
\begin{aligned}
4a_2\beta_3^* &= 2\eta_3 - 1 = -0.76, \qquad \alpha_3^* = 1, \\
4a_2\beta_{1,2}^* &= \tfrac{0.88-0.12}{2} - 1 = -0.62, \\
\alpha_1^* &= \tfrac{0.76}{4\cdot 0.48} \approx 0.396, \qquad \alpha_2^* = \tfrac{0.76}{4\cdot 0.40} = 0.475,
\end{aligned}
$$

yielding $\boldsymbol{\alpha}^* = (0.396, 0.475, 1.000)$, $f = 0.500$, and $g = g^{\max} = 15{,}625\,\mathrm{MW}^2$. The construction is one-shot: the partition is read off directly from $(\eta_{\max}, \eta_{\min})$, no iterative saturation check is needed.

TABLE III
Overbidding Scenario Comparison ($n = 3$, $a_2 = 0.1$, $x_L^o = 400\,\mathrm{MW}$)

Scenario	Ω^c	$4a_2\beta_i^*$	α_1^*	α_2^*	α_3^*	f	$g\,(\mathrm{MW}^2)$	$\sum \Pi_i$ (\$/hr)	loss
S1: original (no overbid)	$\varnothing$	$-2/3$ (sym.)	0.417	0.500	0.625	0.500	22,500	10,800	0%
S2: unclamped sym. ERS	$\{3\}$	$-2/3$ (sym.)	0.347	0.417	1.389	0.500	15,625	10,875	0%
S2: hetero-β (Cor. 5b)	$\{3\}$	$(-0.62, -0.62, -0.76)$	0.396	0.475	1.000	0.500	15,625	10,875	0%
S3: unclamped sym. ERS	$\{2, 3\}$	$-2/3$ (sym.)	0.238	1.111	1.111	0.500	22,500	10,800	0%
S3: hetero-β (Cor. 5b)	$\{2, 3\}$	$(-0.60, -0.70, -0.70)$	0.286	1.000	1.000	0.500	22,500	10,800	0%

S2: x_3^o=30 MW, x_r^o=250 MW; S3: x^o=(210, 45, 45) MW, x_r^o=300 MW; loss = $g^{\max}$ gap.

TABLE IV
ERS Strategy Performance Under Overbidding Scenarios ($n = 3$, $a_2 = 0.1$, $x_L^o = 400\,\mathrm{MW}$)

Scenario	Strategy	$\boldsymbol{\alpha}^*$	f	$g\,(\mathrm{MW}^2)$	$\sum_i \Pi_i$ (\$/hr)
S2 (x_r^o=250 MW)	Coalition	(0.500, 0.500, 0.500)	0.500	15,625	10,875
	RN Nash	(0.521, 0.625, 2.083)	0.750	11,719	10,094
	Pareto-NE	(0.347, 0.417, 1.389)	0.500	15,625	10,875
	Algo. 1 (Cor. 5b)	(0.396, 0.475, 1.000)	0.500	15,625	10,875
S3 (x_r^o=300 MW)	Coalition	(0.500, 0.500, 0.500)	0.500	22,500	10,800
	RN Nash	(0.357, 1.667, 1.667)	0.750	16,875	9,675
	Pareto-NE	(0.238, 1.111, 1.111)	0.500	22,500	10,800
	Algo. 1 (Cor. 5b)	(0.286, 1.000, 1.000)	0.500	22,500	10,800

Pareto-NE bids in italics are infeasible ($\alpha_i > 1$); Algo. 1 enforces feasibility via Cor. 5b.

Table IV summarizes the four primary strategies across Scenarios 2 and 3, confirming that Algorithm 1 matches the Coalition and Pareto-NE aggregate performance while enforcing feasibility. Figure 6 applies all six strategies to Scenario 2 ($x_3^o = 30\,\mathrm{MW}$, $\eta_3 = 0.12$), the same capacity configuration examined analytically above. Under these parameters, $\eta_3 < \frac{1}{2n} = \frac{1}{6}$, so the unconstrained symmetric Pareto bid for Op 3 is $\alpha_3^* = \frac{1}{6\eta_3} \approx 1.39 > 1$ — a feasibility violation. The risk-neutral Nash equilibrium is even more aggressive ($\alpha_3^* \approx 2.08$). Algorithm 1 follows the heterogeneous-β branch of Corollary 5b: it identifies the partition $\Omega = \{1, 2\}$, $\Omega^c = \{3\}$ via max-/min-consensus, recovers $\sum_{j\in\Omega} \eta_j - \sum_{j\in\Omega^c} \eta_j = 0.76$ and $|\Omega| = 2$ from two short auxiliary consensus rounds, and submits (0.396, 0.475, 1.000) — landing the Algo 1 bar exactly at the $\alpha = 1$ boundary while recovering $f = \frac{1}{2}$. Also shown is a heterogeneous-β deviation scenario ($\beta' = [-0.20, -0.15, -0.05]$, "Unclamped OB") and its naive per-operator clamp ("No-overbid"), which can push Op 3 far above 1 without the principled risk re-allocation that Algo 1 performs.

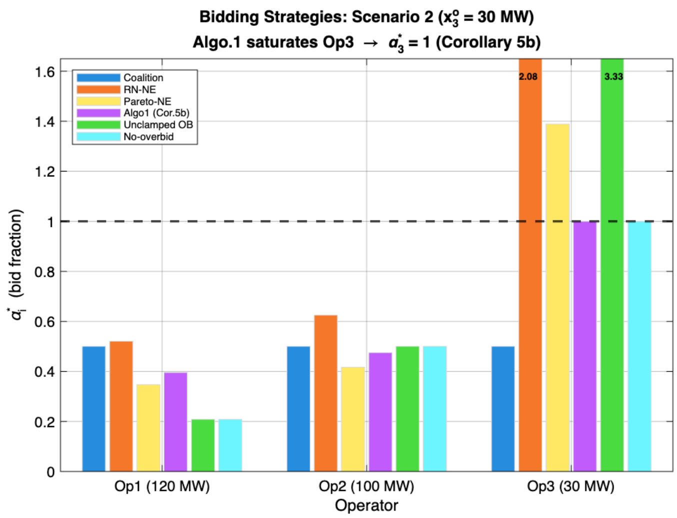

Fig. 6. Bidding strategies α_i^* under Scenario 2 ($x_3^o = 30\,\mathrm{MW}$). The dashed line marks $\alpha = 1$; both the unconstrained Pareto-NE and RN-NE overbid at Op 3, while Algorithm 1 (Corollary 5b saturation branch) places α_3^* exactly at 1. Bars for RN-NE and Unclamped OB at Op 3 extend beyond the y-axis clip; their values are annotated.

Figures 7 and 8 show the profit consequences under Scenario 2. Coalition and Algo 1 achieve the same total profit ($f = 0.5$, $g = g^{\max} = 15{,}625\,\mathrm{MW}^2$), with Op 3 receiving a smaller absolute share due to its reduced capacity. Pareto-NE produces the same aggregate outcome analytically but is infeasible ($\alpha_3^* > 1$). Unclamped overbidding overshoots $f = 0.5$, reducing aggregate efficiency, while the simple per-operator clamp partially recovers it without restoring the Pareto target.

Observation. Overbidding at the ERS optimum is not an artifact of deviating from the Pareto sum condition, but a structural consequence of small renewable share: operator i overbids whenever $\eta_i < \frac{1}{2n}$, regardless of the chosen common β'. No homogeneous-β policy can simultaneously achieve feasibility and $f = \frac{1}{2}$. The heterogeneous allocation of Corollary 5b assigns a more risk-seeking β_i^* to capacity-limited operators (driving $\alpha_i^* = 1$ exactly) and a common, milder β_i^* to the remainder (restoring $f = \frac{1}{2}$), recovering $g^{\max}$ with no efficiency penalty.

E. Constrained Risk-Neutral Nash (Corollary 4)

To isolate the value of risk coordination under capacity constraints, consider Scenario 2 capacities ($x_3^o = 30\,\mathrm{MW}$, $x_r^o = 250\,\mathrm{MW}$) with all operators risk-neutral ($\beta_i = 0$) and overbidding forbidden. Since $\eta_3 = 0.12 < \frac{1}{n+1} = 0.25$, Corollary 4 applies with $\Omega_{\mathrm{RN}} = \{3\}$. Op 3 is clamped to $\alpha_3^* = 1$ and the remaining operators are re-solved:

$$
\alpha_i^* = \frac{1 - \sum_{j\in\Omega_{\mathrm{RN}}} \eta_j}{(n + 1 - |\Omega_{\mathrm{RN}}|)\,\eta_i} = \frac{0.88}{3\eta_i}, \quad i \in \{1, 2\},
$$

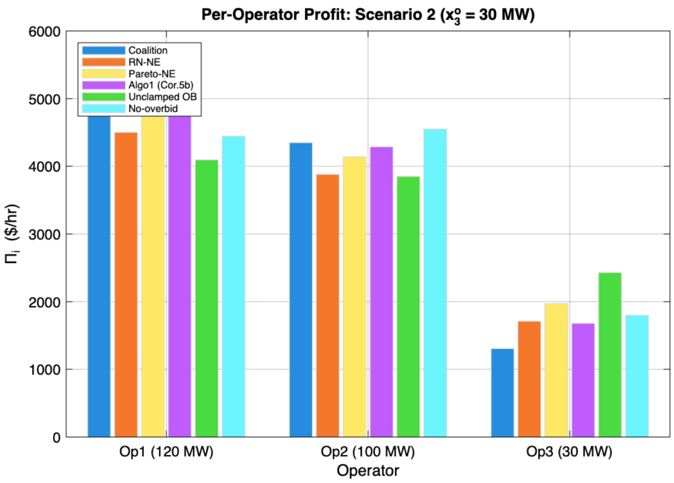

Fig. 7. Per-operator profit Π_i under Scenario 2 for all six strategies. Coalition and Algo 1 achieve the same total profit at $f = 0.5$; risk-neutral and unclamped strategies overshoot $f = \frac{1}{2}$, reducing aggregate efficiency.

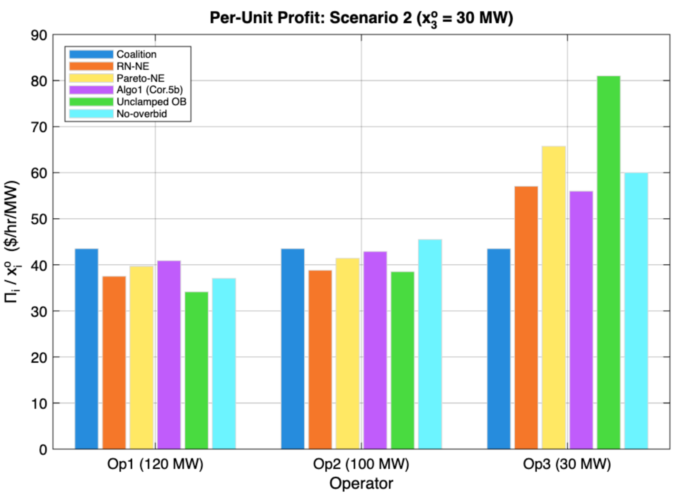

Fig. 8. Per-unit profit Π_i/x_i^o under Scenario 2. Algorithm 1 achieves per-unit equity comparable to the Coalition benchmark; unclamped overbidding distorts Op 3's per-unit revenue upward at the expense of the other operators.

yielding $\boldsymbol{\alpha}^* = (0.611, 0.733, 1.000)$, $f = 0.707$, and $g \approx 12{,}946\,\mathrm{MW}^2$.

Figure 9 compares the three strategies for Scenario 2. The unconstrained RN naturally drives Op 3 to $\alpha_3^* \approx 2.08$ (overbidding); Corollary 4 enforces feasibility by re-solving the equilibrium, but $f = 0.707 \gg \frac{1}{2}$ leaves market efficiency at only $\approx$83% of $g^{\max} = 15{,}625\,\mathrm{MW}^2$. This quantifies the efficiency cost of foregoing risk coordination: even when overbidding is eliminated, risk-neutral operation incurs a persistent market efficiency loss that risk-seeking coordination avoids.

F. Multiple Operator Saturation (Corollary 5b)

To validate the generality of Corollary 5b beyond $|\Omega^c| = 1$, we introduce Scenario 3 by setting $x^o =$

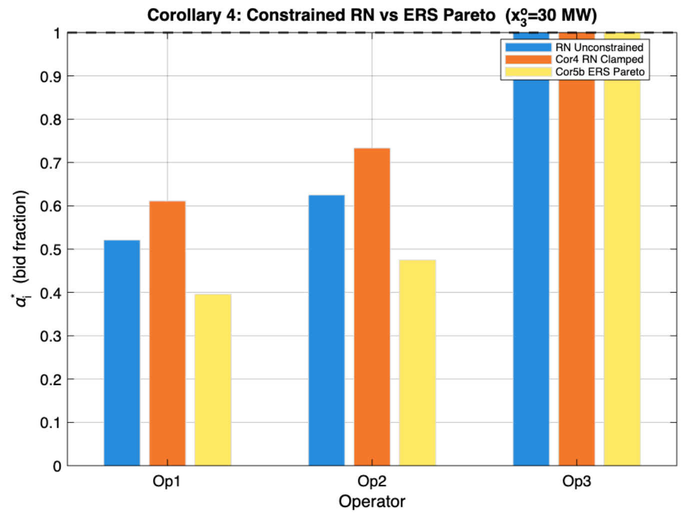

Fig. 9. Corollary 4: bidding strategies under unconstrained RN and constrained RN (Cor. 4 clamped) for Scenario 2 ($x_3^o = 30\,\mathrm{MW}$). Constrained RN enforces feasibility but leaves $f = 0.707 \gg \frac{1}{2}$, incurring a persistent efficiency loss relative to risk-seeking coordination.

$(210, 45, 45)\,\mathrm{MW}$ giving $\eta = (0.70, 0.15, 0.15)$. The unconstrained ERS formula yields $\alpha_{2,3}^* \approx 1.11 > 1$, so Ops 2 and 3 would overbid; the heterogeneous-β construction of Corollary 5b corrects this.

With $\eta_{\max} = 0.70$ and $\eta_{\min} = 0.15$, the cutoff and partition are

$$\eta_c = \tfrac{1}{2}(0.70 + 0.15) = 0.425, \qquad \Omega = \{1\}, \quad \Omega^c = \{2, 3\}.$$

We have $\sum_{j\in\Omega} \eta_j = 0.70$, $\sum_{j\in\Omega^c} \eta_j = 0.30$, so $D \triangleq \sum_{j\in\Omega} \eta_j - \sum_{j\in\Omega^c} \eta_j = 0.40$ and $|\Omega| = 1$. Substituting into (36):

$$\alpha_1^* = \frac{D}{2\,|\Omega|\,\eta_1} = \frac{0.40}{2 \times 0.70} = \frac{2}{7} \approx 0.286, \qquad \alpha_2^* = \alpha_3^* = 1.$$

One can verify $f = \frac{2}{7} \times 0.70 + 1 \times 0.15 + 1 \times 0.15 = 0.500$ and $g = g_{\max} = 22{,}500\,\mathrm{MW}^2$, achieving the full Pareto optimum even with two saturating operators. This contrasts sharply with a naïve homogeneous-β_0 approach, which would give $f = 0.825$ and $g \approx 13{,}001\,\mathrm{MW}^2$ — confirming that the heterogeneous construction eliminates the efficiency penalty regardless of $|\Omega^c|$.

Figure 10 compares all six strategies under Scenario 3. Both RN-NE and the unconstrained Pareto-NE overbid at Ops 2 and 3, while Algorithm 1 (Corollary 5b) sets $\alpha_{2,3}^* = 1$ exactly and recovers $f = \frac{1}{2}$ without the efficiency penalty incurred by naive clamping. Figures 11 and 12 show the corresponding per-operator and per-unit profits.

G. Solutions under Distributed Algorithms

Figures 13–14 show convergence of the consensus states. The initial conditions follow Algorithm 1: the utility sets $\xi_4(0) = 1$, $\psi_4(0) = 0$; each operator sets $\xi_i(0) = 0$, $\psi_i(0) = x_i^o$. All agents converge to the common average $\xi_i(k) \to \frac{1}{4}$ and $\psi_i(k) \to \frac{x_r^o}{4} = 75\,\mathrm{MW}$. From these converged values,

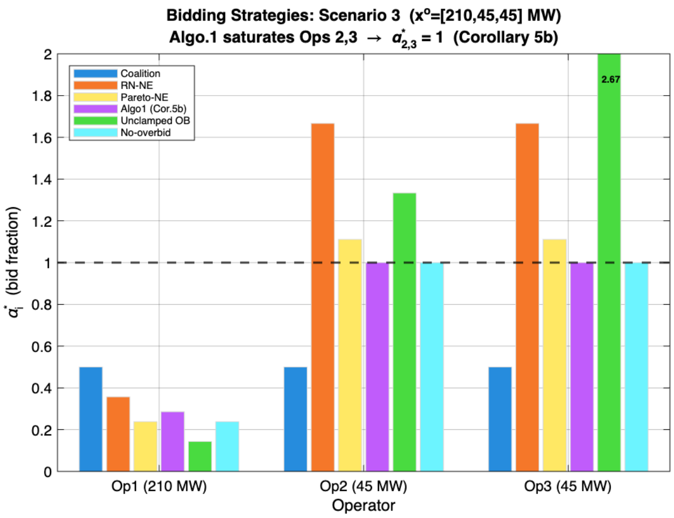

Fig. 10. Bidding strategies under Scenario 3 ($x^o = (210, 45, 45)$ MW). Both RN-NE and Pareto-NE overbid at Ops 2 and 3; Algorithm 1 (Cor. 5b) places $\alpha^*_{2,3}$ exactly at 1. Naive clamping reduces f below $\frac{1}{2}$.

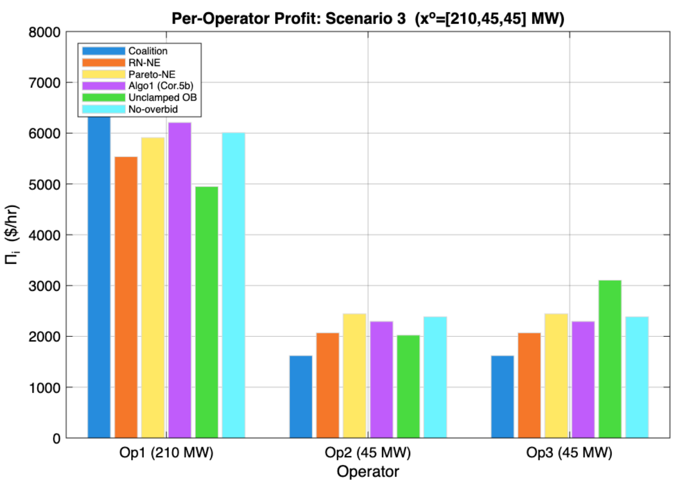

Fig. 11. Per-operator profit Π_i under Scenario 3 for all six strategies. Coalition and Algo 1 both achieve $\sum \Pi_i = 10{,}800$ \$/hr at $f = 0.5$; risk-neutral and unclamped strategies reduce aggregate efficiency.

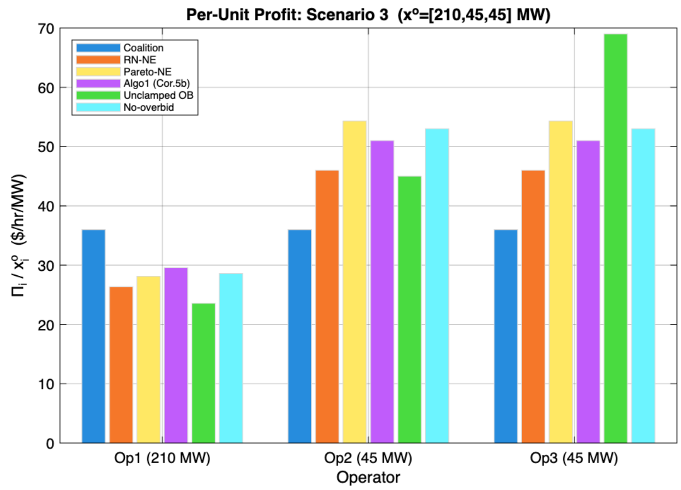

Fig. 12. Per-unit profit Π_i/x_i^o under Scenario 3. Algorithm 1 restores per-unit equity comparable to the Coalition benchmark.

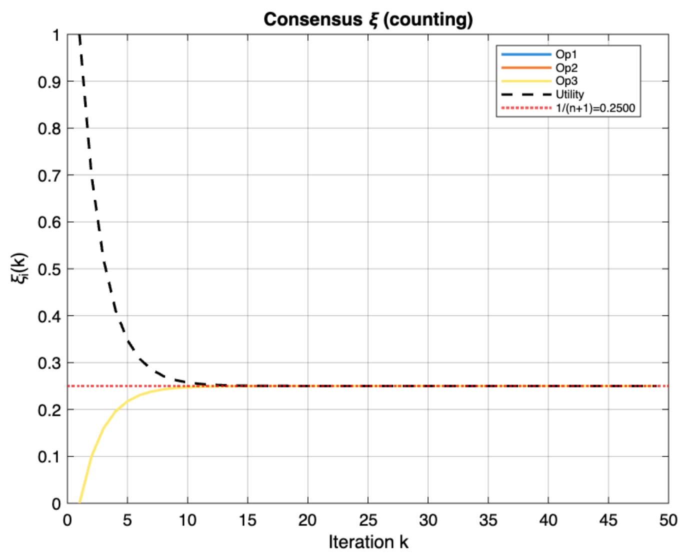

Fig. 13. Algorithm 1 consensus trajectory for ξ_i.

each operator i locally computes the quantities shown in Table V, confirming that every operator independently recovers the correct $\hat{n}_i = 3$, η_i, and $\beta'_i = -\frac{1}{6}$ without any additional communication.

TABLE V
Quantities computed locally by each operator after consensus convergence

Op.	ξ_i^∞	$\hat{n}_i$	η_i	β'_i
1	0.25	3	2/5	−1/6
2	0.25	3	1/3	−1/6
3	0.25	3	4/15	−1/6

Remark 3. The convergence rate of Algorithm 1 on the star graph $K_{1,n}$ is $O(\log(1/\varepsilon))$ per consensus round, independent of the number of operators n. This follows from the algebraic connectivity $\lambda_2(L) = 1$ of $K_{1,n}$, which is established in Section V and does not degrade as n grows.

The Corollary 5b branch of Algorithm 1 is validated on Scenarios 2 and 3. In each case operators first run max- and min-consensus on η_i to recover $\eta_{\max}$ and $\eta_{\min}$ (Phase 3a), then each operator locally computes $\eta_c = (\eta_{\max} + \eta_{\min})/2$ and determines its partition membership. Operators not in Ω_{ERS} then execute Phase 4: a second ξ-protocol seeded with $\xi_i^{(2)}(0) = 1$ for $i \notin \Omega_{\text{ERS}}$ and 0 otherwise, recovering $n - |\Omega_{\text{ERS}}| = \xi_i^{(2),\infty}/\xi_i^\infty$; and a ψ-protocol seeded with $\psi_i^{(2)}(0) = \eta_i$ for $i \notin \Omega_{\text{ERS}}$ and 0 otherwise, recovering $\sum_{j \notin \Omega_{\text{ERS}}} \eta_j = \psi_i^{(2),\infty}/\xi_i^\infty$. In Scenario 2 ($\eta_c = 0.30$): Ops 1 and 2 are not in Ω_{ERS} and seed the Phase 4 protocols, while Op 3 ($\in \Omega_{\text{ERS}}$) sets $\alpha^*_3 = 1$ directly. In Scenario 3 ($\eta_c = 0.425$):

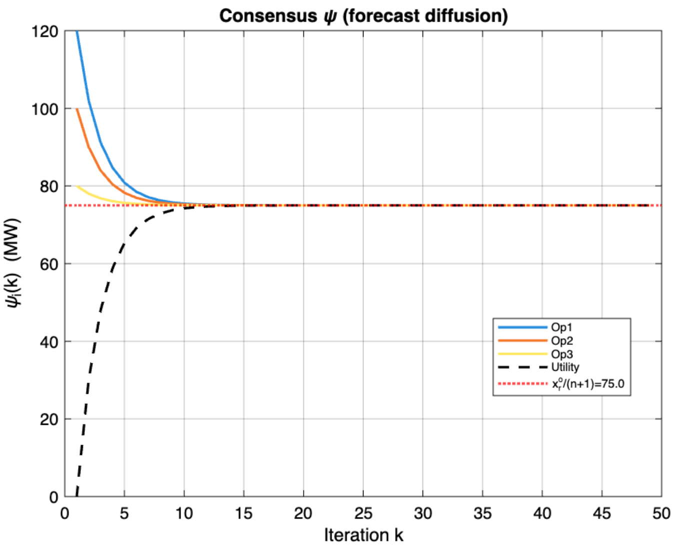

Fig. 14. Algorithm 1 consensus trajectory for ψ_i.

Op 1 alone is not in Ω_{ERS} and executes Phase 4, while Ops 2 and 3 ($\in \Omega_{\mathrm{ERS}}$) saturate at $\alpha^*_{2,3} = 1$. In both cases the Phase 4 consensus rounds converge and the locally recovered $(n - |\Omega_{\mathrm{ERS}}|)$ and $\sum_{j \notin \Omega_{\mathrm{ERS}}} \eta_j$ yield bids $\boldsymbol{\alpha}^*$ indistinguishable from the centralized Corollary 5b solution, confirming that partition information is correctly propagated without any central coordinator.

H. Scalability Discussion

The numerical example above uses $n = 3$ operators to keep the exposition concrete. The theoretical results, however, are fully general: Theorems 1–2 and Corollaries 1–5b hold for any $n \geq 1$, and Algorithm 1 requires no modification as n grows. Three structural properties support scalability. First, from Remark 3, the convergence rate of Algorithm 1 is $O(\log(1/\varepsilon))$ per consensus round on the star graph, independent of n. Second, each operator performs only local computations (a single division and two multiplications after consensus), so per-operator complexity does not grow with n. Third, the overbidding threshold $\eta_i < 1/(2n)$ becomes more stringent as n increases, so in large markets with many similarly-sized operators the unconstrained formula (35) applies without any clamping. Simulation studies at larger scale ($n \in \{10, 50\}$) are deferred to future work, where they will also address sensitivity of the Pareto condition to heterogeneous capacity distributions and transient operator participation.

VII. Conclusion

This paper presented a risk-aware bidding framework in which the risk sensitivity parameter β_i acts as a coordination variable for renewable generation operators in electricity markets. We showed that the noncooperative game admits structured Nash equilibria and that, by appropriate risk allocation, these equilibria can recover Pareto-efficient aggregate outcomes. In particular, the analysis established explicit relationships between risk parameters, bidding coefficients, and market efficiency.

A distributed consensus-based implementation was developed for the one-shot strategy, allowing each operator to estimate required aggregate quantities using only local communication. The illustrative example confirmed that the distributed Algorithm 1 reproduces the targeted bidding behavior without centralized disclosure of private forecasts or cost data. When the no-overbidding constraint is active, a minor extension using a third consensus protocol allows each operator to determine its constrained bid locally after recovering the aggregate saturation quantities $\sum_{j \in \Omega_{\mathrm{ERS}}} \eta_j$ and $\sum_{j \notin \Omega_{\mathrm{ERS}}} \eta_j$ through neighborhood communication alone.

A detailed overbidding analysis established per-operator saturation thresholds from (34) and showed that operators with smaller capacity shares saturate first under a common risk parameter. The individual profit analysis revealed that the no-overbidding projection shifts $f(\boldsymbol{\alpha})$ below $\frac{1}{2}$ and reduces the payoff of all operators through the shared market-clearing term, not only those whose bids are clamped. This individual-level effect, quantified via (5b) and illustrated numerically in the constrained Scenario 2 study, demonstrates that the performance impact of feasibility constraints is collective rather than local.

Overall, the results demonstrate that risk-aware design can align self-interested bidding with improved collective efficiency while preserving decentralization and privacy. Future work will incorporate network congestion and locational pricing, extend to non-Gaussian uncertainty structures and time-varying operator participation, and analyze robustness under communication delay, packet loss, and adversarial behavior.

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